

Image-Based Defect Detection Robustness

Report

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Introduction

The program designed by ImageDefect Team 8 integrates an AI classifier into a classical detection model, creating a hybrid model to detect defects in wood. The program relies on the AI to correctly determine a Pass/Fail, while using the classical methods to determine causality or possibly override the AI. This report focuses on the robustness of the program under altered images.

1. Approach & Methodology

Alterations to images were made via Windows' native 'Photos' app, with basic usage of rotations, adjustments, filters, and markings.

1.1 Alterations

There were 5 primary alterations done to every image: rotation of 90, 180, or 270 degrees, light adjustment, color adjustment, application of filters, and marking.

Rotation

Rotations were done in multiples of 90 degrees, causing a sideways image, flipped image, or upside-down image.

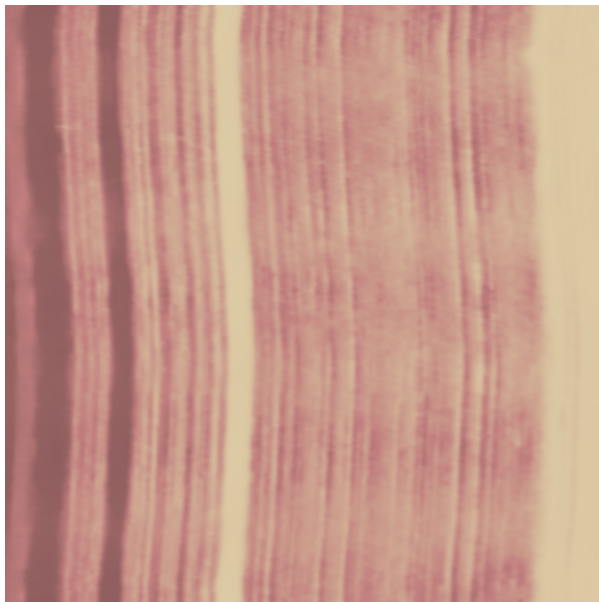


Figure 1: Regular Image

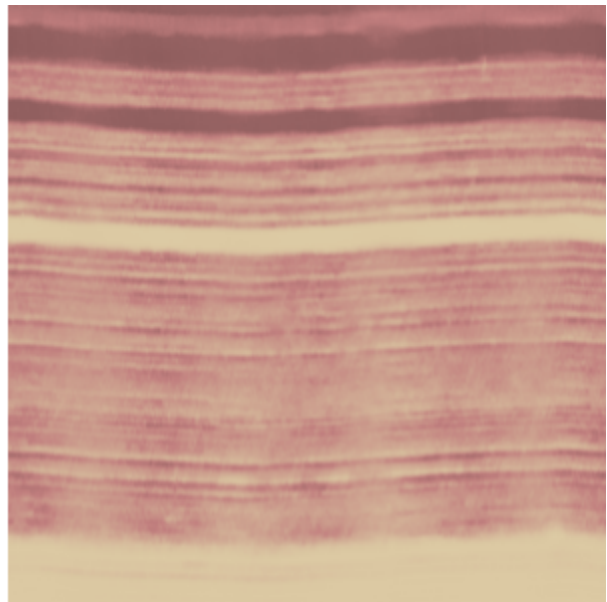


Figure 2: 90 Degree Rotation (Sideways)

Light Adjustment

Light adjustment allows adjustments of these image values: Brightness, exposure, contrast, highlights, shadows, or vignette. These adjustments were used to darken and brighten images

Color Adjustment

Color adjustment allows adjustments of these image values: Saturation, warmth, tint, and sharpness. Saturation, warmth, and tint were used to make color changes, while sharpness dictated blur.

Filter

Filters could be applied to the image under these available filters with different intensities:

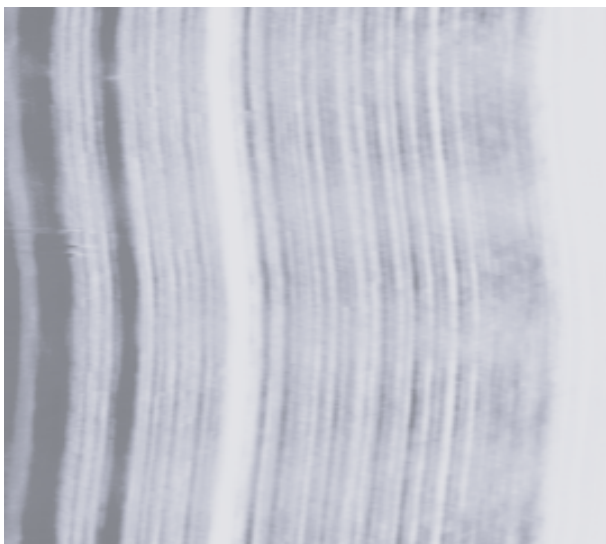
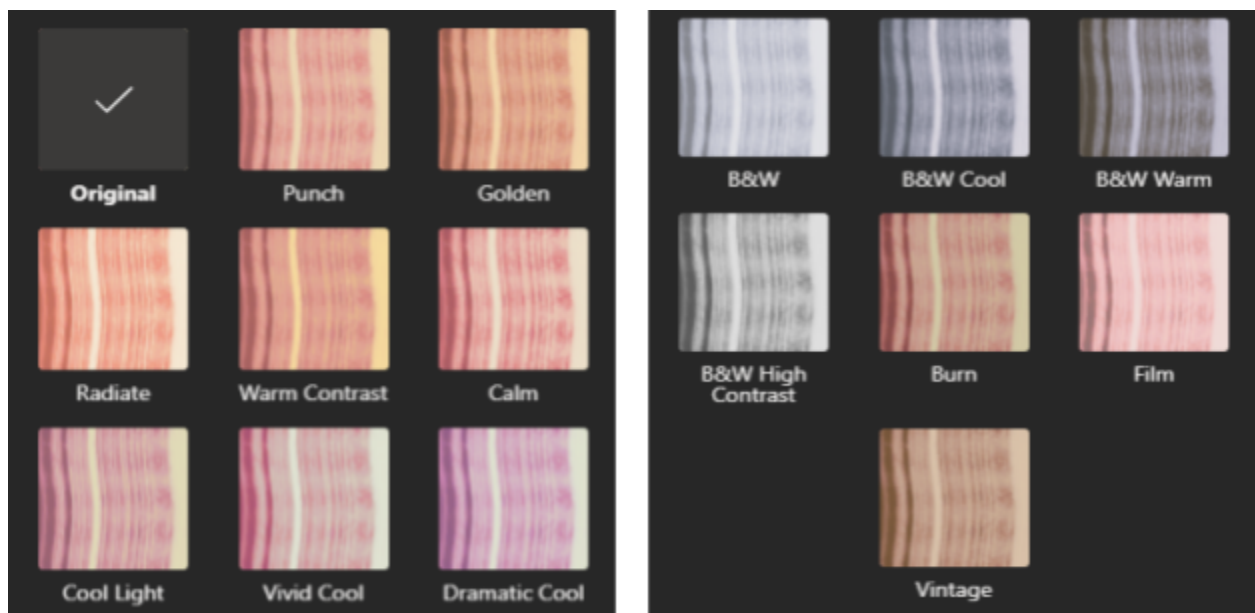


Figure 3: Black and White (0) Filter

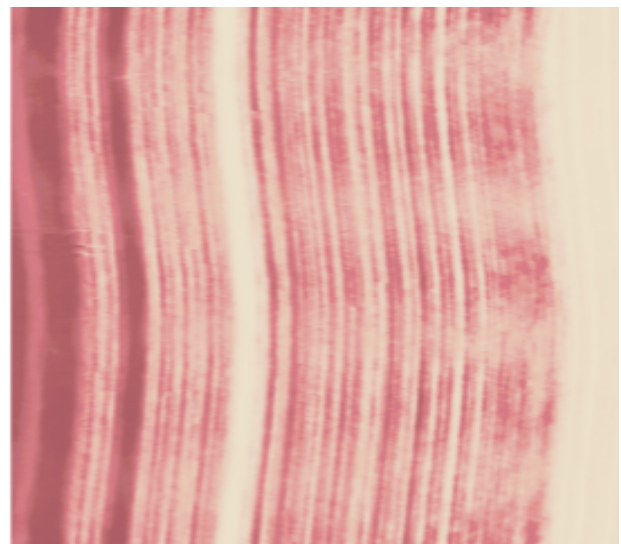


Figure 4: Calm (100) Filter

Mark

Marks can be made on the image, creating a 'scratch' that can change the results of testing

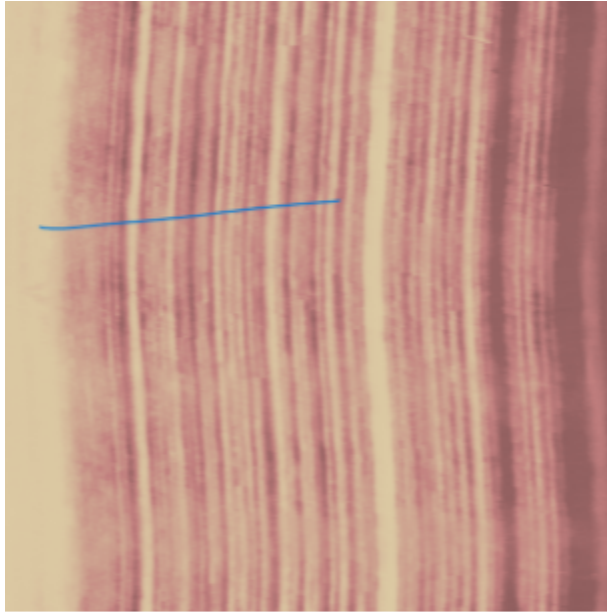


Figure 5: Added Line

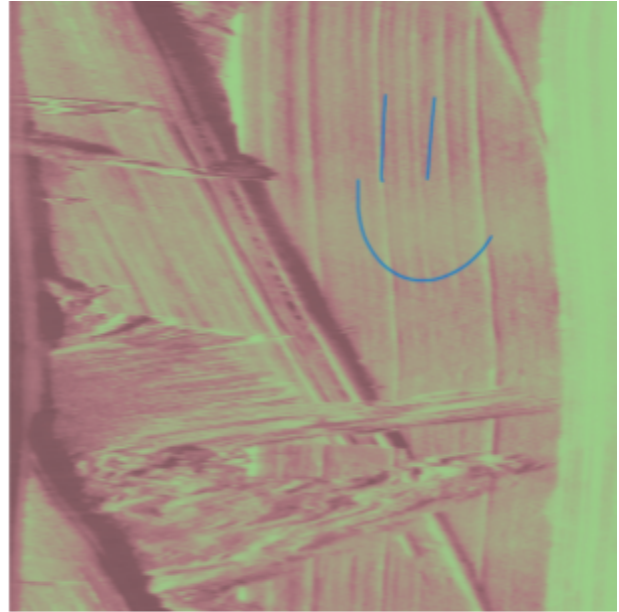


Figure 6: Added Smile

1.2 Batch & Control Testing

To test the robustness of the program, 14 images were selected, 7 from passing and 7 from failing images. All these images were then altered using the above-listed methods and run through the program immediately after the original image was tested. Pass/Fail of the original images was recorded, while the altered images had comprehensive results issued.

2. Results & Analysis

This section contains in-depth analysis of the results, as well as the raw data recorded during the testing process.

2.1 Results

Image	Original	Alteration	New Result	Match	Notes
Image	Original Score	What was edited in the picture	Pass (75%) Disagreement	Yes or No	Any particular details noticed

1	Pass	Marking in a different color	Pass (75%) Disagreement	Yes	Line marked by detection model, overridden by AI
2	Pass	Rotation, Black & White Filter	Fail (96%)	No	Entire image was flagged as a failure point
3	Pass	Rotation, color alteration, blur	Fail (82%) Disagreement	No	No flags; AI failed image while detection passed
4	Pass	Markings with natural coloration, hue shift, desaturation	Fail (100%)	No	Both AI and detection model failed the image
5	Pass	Overexposed and increased contrast	Fail (99%) Disagreement	No	AI flagged image, detection model passed
6	Pass	Rotation, blur, pink	Fail (100%)	No	Horizontal grain flagged as defects
7	Pass	Rotation, vintage filter	Fail (86%)	No	Horizontal grain flagged as defects
8	Fail	Rotation, desaturation, blur	Fail (76%)	Yes	Correct markings flagged
9	Fail	Hue shift, added smiley face	Fail (80%)	Yes	Marked correct areas and non-grain-aligned parts of the smiley face
10	Fail	Overexposed, pink, blur	Fail (100%)	Yes	AI failed, passed by detection model, AI override
11	Fail	Rotation, underexposed and darkened	Pass (91%)	No	Both AI and detection model could not detect defects
12	Fail	Rotation, high contrast	Fail (91%)	Yes	Horizontal grain flagged as defects
13	Fail	Rotation, high shadow, lower white, blur	Fail (92%)	Yes	Horizontal grain and brushes flagged as defects
14	Fail	Rotation, vintage filter, blur, darkened	Fail (80%)	Yes	All appropriate brushes detected

2.2 Analysis & Interpretation

Out of the 14 altered images, 7 of the altered images returned different results from the original images, giving a 50% consistency rate between the original and altered images. Of the 7 'Passes', 6 were flagged as failures after alteration, while only 1 failed image returned as a pass. This means that the system is much more likely to produce a false positive, detect a defect when there isn't one, than to pass defects. This is good for a material detection system, where any faulty parts may have catastrophic consequences. In this way, it is more worthwhile to flag perfect and pristine material.

Of the clean images that were flagged, 57% had been rotated in some form, and 75% of those rotated images had highlighted the entirety of the grain as being a defect. This likely implies that the program recognizes anything horizontal as a scratch or brush, rather than a possibility of the grain being horizontal instead of vertical. This is again seen in images 12 and 13, which highlighted horizontal grain as defects.

3. Improvements

After reviewing the results of this testing, some improvements can be made to the program. The best improvement I would suggest to the system is having a non-static grain direction. The current system assumes all grain to be vertical, meaning any horizontal or diagonal grain is recognized as a scratch or brush, rather than filtering it out.

Conclusion & Final Thoughts

The robustness test of the program produced very interesting results, allowing the team to see the limitations of the program and the possible improvements that could be made. Thorough testing was an interesting experience, and modifying images for testing was a unique experience.